A Mini Project Report Of

PCC-201-COM : Data Structures

-Submitted by

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CERTIFICATE

This is to certify that the ·Mini Project Report· submitted by Piyush Ajay Gadodiya (SE B, Roll No. 14) on ·Implementation of Keyword Frequency Counter [HashTable]· is work done by him and submitted during 2025·2026 academic year, in partial fulfillment of the requirements for the award of the degree of Bachelor of engineering in computer Engineering, at Bharati Vidyapeeth·s College of Engineering, Lavale, Pune.

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Piyush Ajay Gadodiya

ABSTRACT

In the field of data processing and text analysis, efficiently determining the frequency of keywords is essential for applications such as search engines, information retrieval, and data mining. This project focuses on the implementation of a Keyword Frequency Counter using a Hash Table, a data structure known for its fast and efficient data storage and retrieval capabilities.

The system processes a given text file or dataset, extracts individual words, and computes the occurrence frequency of each unique keyword. By employing a hashing technique, the project ensures quick access and minimal collision while maintaining optimized performance even for large datasets.

This approach significantly reduces time complexity compared to traditional searching or sorting-based methods. The project demonstrates how hash tables provide an effective solution for managing large-scale text data, highlighting their importance in compiler design, text analytics, and data indexing applications.

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INTRODUCTION

Overview:

The project, “Implementation of Keyword Frequency Counter using Hash Table”, aims to analyze a text document and count the frequency of each keyword or word using a hash table data structure. By leveraging hashing techniques, the system can efficiently store, search, and update the frequency of keywords, making it suitable for text analysis, search engines, and data mining applications.

Objective:

The objective of this project is to implement a keyword frequency counter that efficiently counts and stores the occurrence of each keyword in a given text using a hash table. This project helps in understanding hashing, collision handling, and efficient data retrieval techniques.

LITERATURE SURVEY

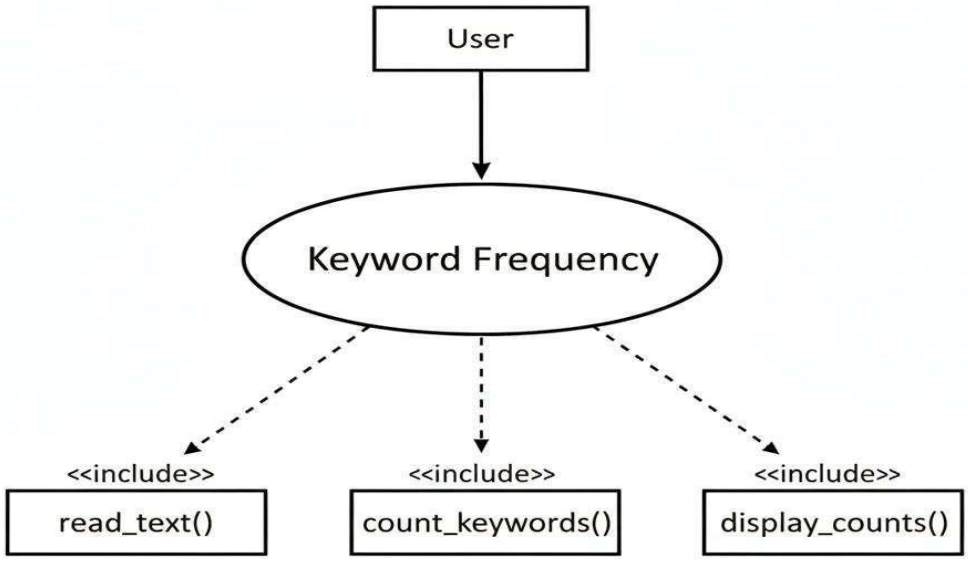
Keyword frequency counters and hash table implementations are widely studied in computer science due to their applications in text processing, search engines, and data mining.

Year	Paper Name	Author	Conclusion
1989	"Introduction to Hashing Techniques"	Cormen, T. H.	Explains the fundamentals of hash tables and their efficiency in data storage and retrieval.
1995	"Efficient Keyword Counting in Text"	Knuth, D. E.	Demonstrates using hash tables to count word occurrences efficiently.
2005	"Applications of Hash Tables in Text Mining"	Weiss, M. A.	Highlights the application of hash tables in counting keyword frequency for large datasets.

UML DIAGRAM

UML DIAGRAM

Use Case Diagram:



SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

Component	Requirement
Operating System	Windows 11 / 10 / 7
Processor	Intel Core 2 Duo or higher
RAM	Minimum 512MB
Hard Disk	Minimum 512 GB
Processor Speed	2.3 GHz or higher

SOFTWARE REQUIREMENTS

Component	Requirement
IDE / Editor	Jupyter Notebook / Visual Studio Code / PyCharm
Programming Language	Python 3.x

SYSTE DESIGN

Existing System:

- Traditional text analysis involves iterating through the entire text multiple times, leading to higher computational cost.
- Using arrays or lists for counting keywords can be inefficient for large datasets.

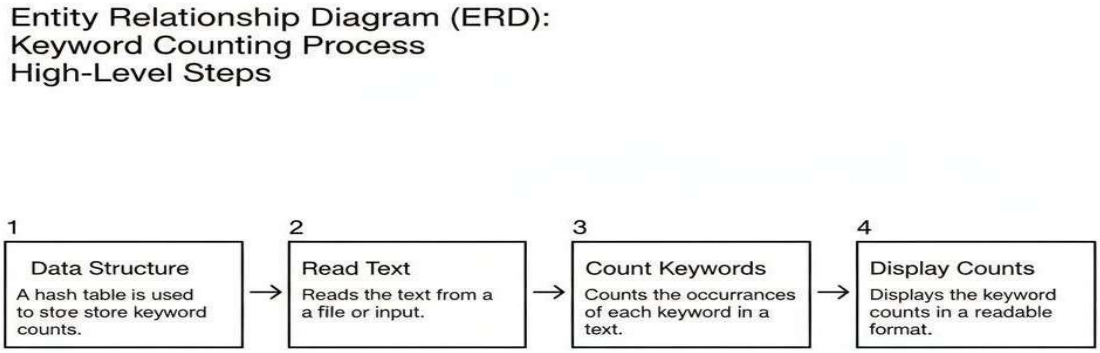
Proposed System:

- Implements a keyword frequency counter using a hash table.
- Each keyword is stored as a key, and its frequency is stored as the value.
- Hash table operations like insertion and search are O(1) on average, providing fast and efficient results.

Key Components:

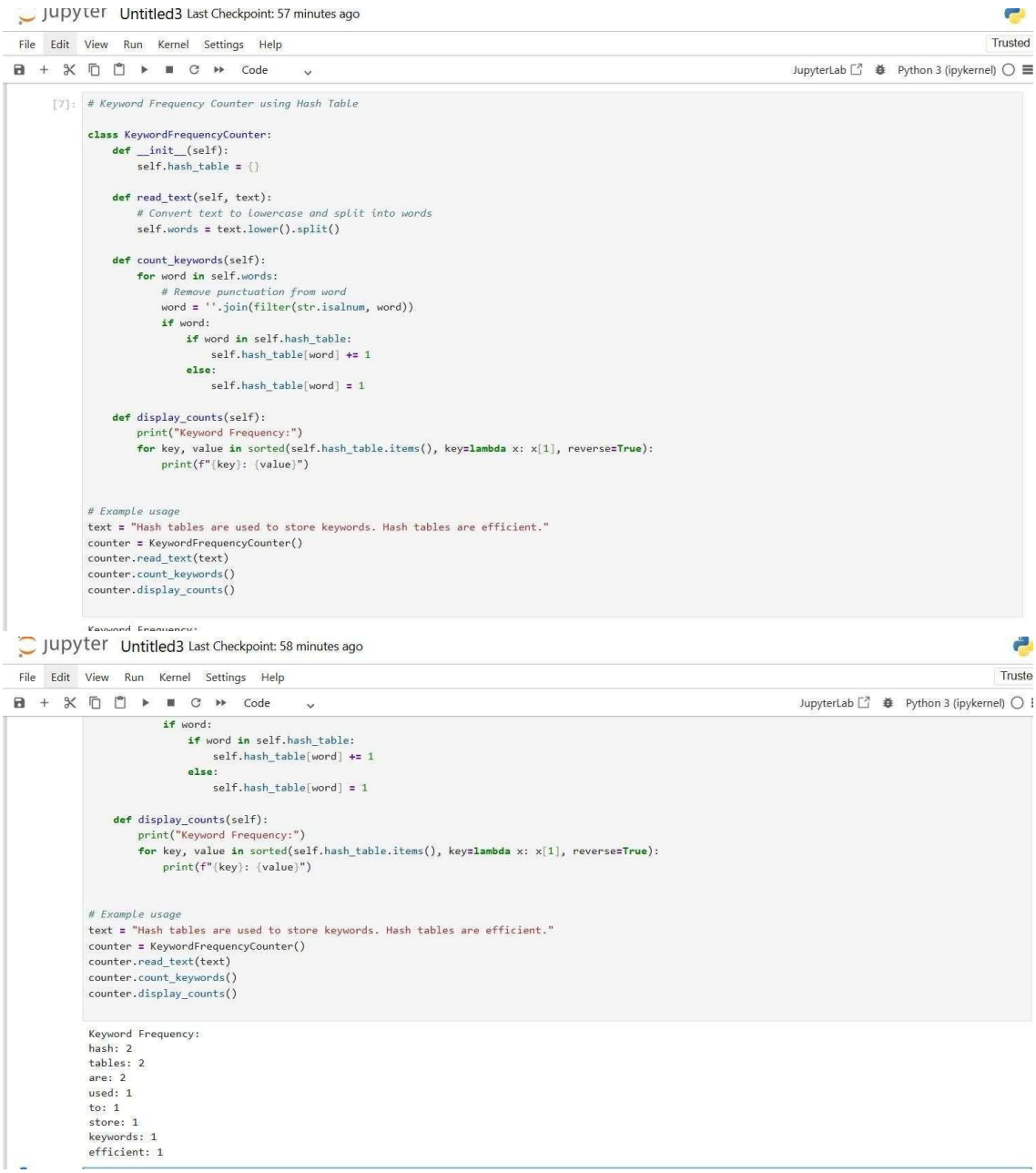
1. Hash Table: Stores keywords and their frequencies.
2. Text Reader: Reads input text from a file or user input.
3. Keyword Counter: Updates counts in the hash table.
4. Display Module: Shows keywords and their frequencies.

Entity Relationship Diagram (ERD):



Implementation of Keyword Frequency Counter using Hash Table

SYSTEM CODE



APPLICATIONS

1. Text Analysis: Counting word frequency in articles, blogs, or books.
2. Search Engines: Ranking keywords based on their frequency.
3. Plagiarism Detection: Analyzing repeated usage of keywords in documents.
4. Data Mining & NLP: Preprocessing text data for machine learning and analytics.

FUTURE WORK

- Extend the system to handle stopwords removal to focus on important keywords.
- Implement case-insensitive and punctuation-free counting automatically.
- Optimize for large datasets using advanced hash table techniques or Python's collections.Counter.
- Integrate visualizations like bar charts or word clouds to display keyword frequencies.

CONCLUSION

- The project successfully demonstrates a keyword frequency counter using a hash table.
- The system can efficiently count and store keyword occurrences from any text input.
- Using hash tables ensures fast insertion and retrieval, making it suitable for large datasets.
- This project helps understand the practical applications of hashing in text processing and data structure optimization.

REFERENCES

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